

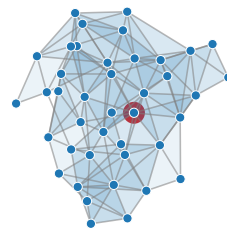
a 13 Å neighborhood



b Alpha complex: 3 Å



c Alpha complex: 4 Å



d Compatible restrictions

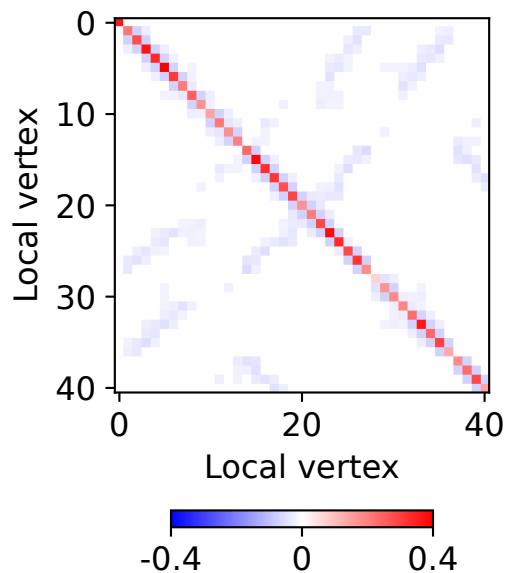
$$q_c = 0, \quad q_v = 1 \ (v \neq c)$$

$$\rho_{\sigma, \tau} = \frac{F(\sigma)}{F(\tau)} \prod_{v \in \tau \setminus \sigma} q_v$$

$$D^1 D^0 = 0$$

$$F(v) = 1, \quad F(ij) = d_{ij}$$

e Center-zero operator



f Spectral descriptors

